

DO POORER INDIAN
STATES GROW FASTER?

AN EVIDENCE OF CONVERGENCE

SOLOW MODEL : BASIC IDEA

01

Solow Growth Model explains growth via capital, population, and technology.

02

Due to diminishing returns to capital, economies tend to slow down as they accumulate more capital.

03

Economies converge to a steady state where capital and output per worker are constant.

CONVERGENCE HYPOTHESIS

Poor states (low initial income) are far from their steady state
higher marginal productivity of capital
Thus, faster growth.

Rich states (high initial income) are closer to their steady state.
Thus, slower growth.

Poor economies grow faster than rich ones.
Growth rate is negatively related to initial income.
This is called β -convergence.

TYPES OF CONVERGENCE

Absolute Convergence

Same steady state for all economies
Poor states catch up automatically

Conditional Convergence

Different steady state for all economies, based on savings, population and technology.

Because Indian states differ in structural factors like investment, population growth, and technology, they converge to different steady states, making conditional convergence more appropriate.

DATASET OVERVIEW

- **Data Structure:** Panel Data (Longitudinal) covering multiple states over a 14-year period.
- **Geographical Scope:** 12 Indian States
- **Includes:** Assam, Gujarat, Himachal Pradesh, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Odisha, Punjab, Tamil Nadu, Uttar Pradesh, and West Bengal.
- **Time Period:** 2011 – 2024 (Annual data).
- **Sample Size:** 168 total observations (12 states × 14 years).
- **Key Variables Categories:**
 1. Economic Indicators: NSDP Per Capita (constant price)
 2. The Technology proxies: Number of Telephone per 100 population
Credit–Deposit Ratio of Scheduled Commercial Banks
Availability of Power (net crore units)
 3. Human capital proxy: Human Development Index

DATA

Source: Handbook of Statistics on Indian States, Reserve Bank of India (RBI).

Global data lab: subnational

Construction of Technology Index using PCA

The Technology Index is constructed using the following indicators:

- Number of Telephone per 100 population
- Credit–Deposit Ratio of Scheduled Commercial Banks
- Availability of Power (net crore units)

These variables capture infrastructure, financial access, and energy availability, which are key components of technological development.

```
. pca Telephone Bank Power, correlation
```

```
Principal components/correlation      Number of obs   =      168
                                      Number of comp.   =       3
                                      Trace               =       3
Rotation: (unrotated = principal)     Rho              =     1.0000
```

Component	Eigenvalue	Difference	Proportion	Cumulative
Comp1	1.64916	.631406	0.5497	0.5497
Comp2	1.01775	.684661	0.3393	0.8890
Comp3	.333091	.	0.1110	1.0000

```
Principal components (eigenvectors)
```

Variable	Comp1	Comp2	Comp3	Unexplained
Telephone	0.2224	0.9375	0.2675	0
Bank	0.7115	0.0315	-0.7020	0
Power	0.6666	-0.3465	0.6601	0

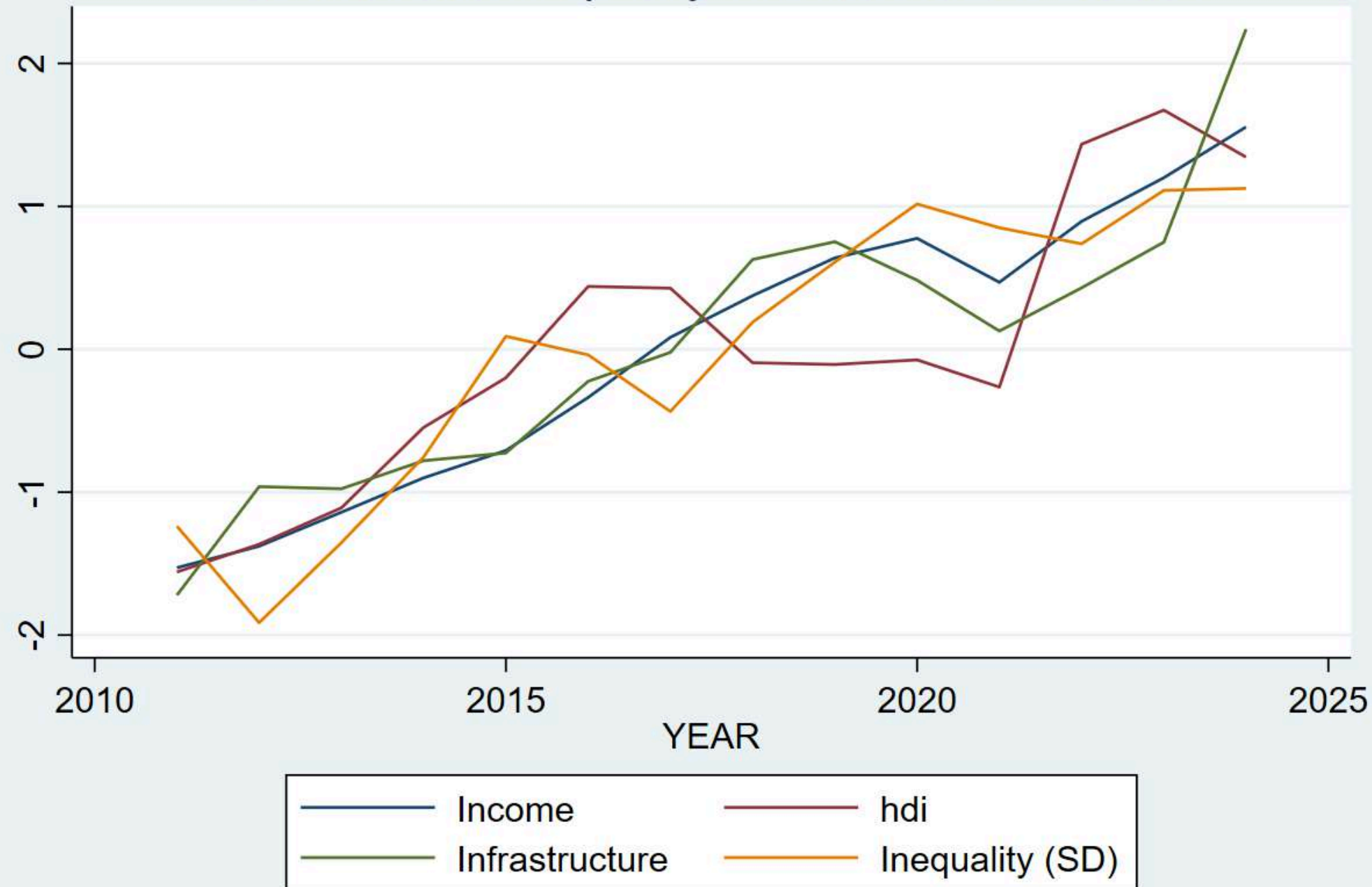
```
. predict index1
```


DESCRIPTIVE STATISTICS

Variable	Obs	Mean	Std. dev.	Min
Max				
nsdppercap~a	168	96350.08	41701.07	31851.74
202064.6				
hdi	168	.7191107	.0620933	.599
.86				
telephone	168	89.94464	26.25727	39
174.46				
Bank	168	63.63929	22.76036	29.7
123.3				
Power	168	6333.72	4463.889	95
20693				
index1	168	-1.67e-09	1.284195	-2.150144
3.311465				
y	168	11.37212	.4723756	10.36885
12.21634				

- Here, $y = \ln(\text{nsdppercapita})$
- Mean per capita NSDP: ₹96,350 with high SD (₹41,701) - large interstate inequality
- HDI ranges from 0.599 to 0.86 : significant human development disparity
- index1 is mean-zero (as expected from PCA, mean = $-1.67e-09$)
- Log income (y) ranges from 10.37 to 12.22 (i.e., ₹31,920 to ₹2,02,540)

Growth and Inequality Trends Across States



ANALYSIS OF ORIGINAL DATA (WITHOUT STANDARDIZATION)

- Mean y rises from 11.07 (2011) to 11.68 (2024) : steady income growth
- HDI rises from 0.687 to 0.747 : improving human development
- index1 rises from -0.38 to +0.49 : improving infrastructure
- SD of y stays around 0.42–0.47 : inequality persisting (relevant for convergence analysis)

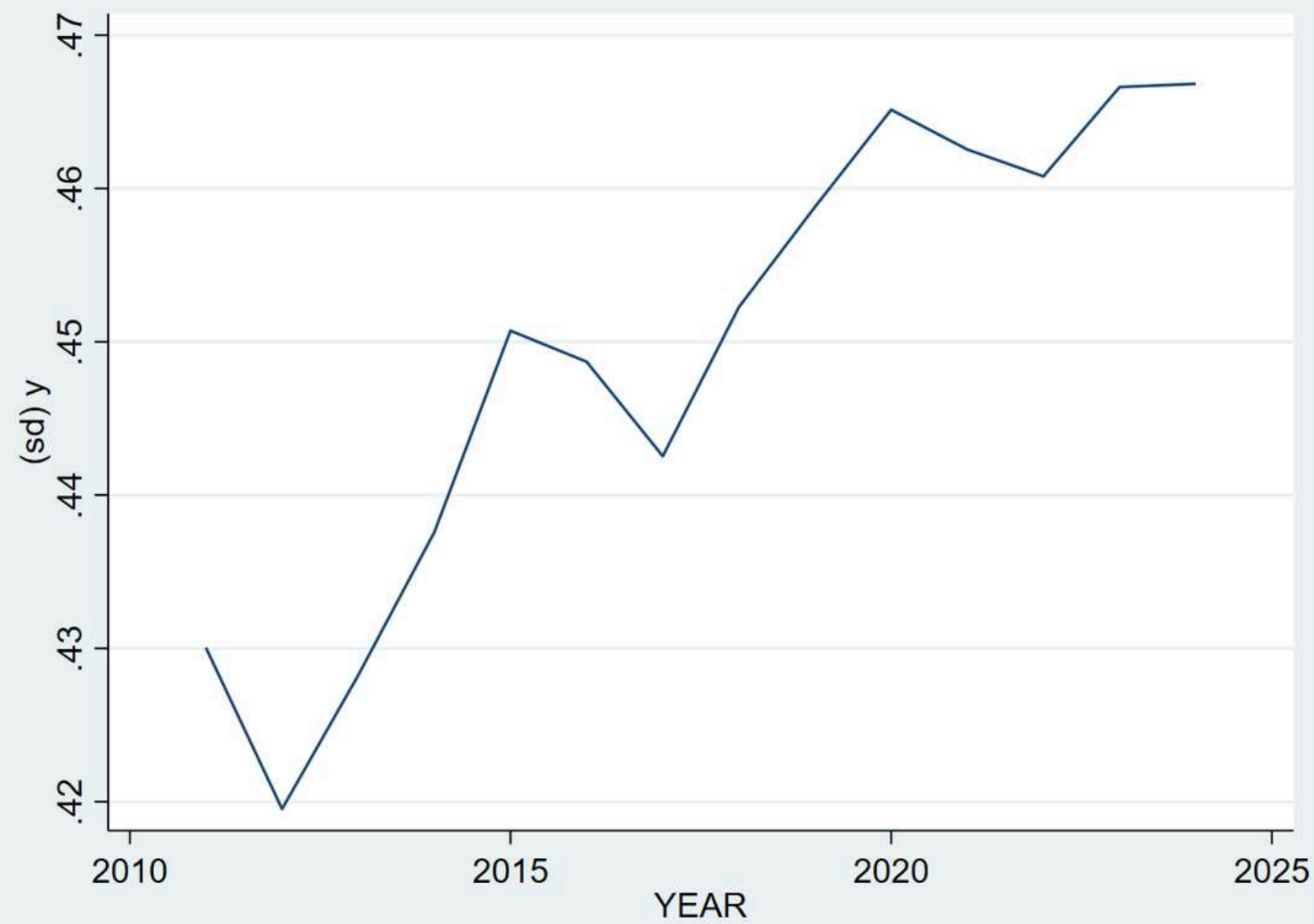
standardization of all variables using Z-score

Sigma (σ) Convergence

- σ -convergence measures dispersion of income across regions over time
- Typically measured using:
 - Standard deviation (SD) of log income
 - Coefficient of variation (CV)
- Condition: Decreasing dispersion (σ -convergence)
- Increasing dispersion (σ -divergence)

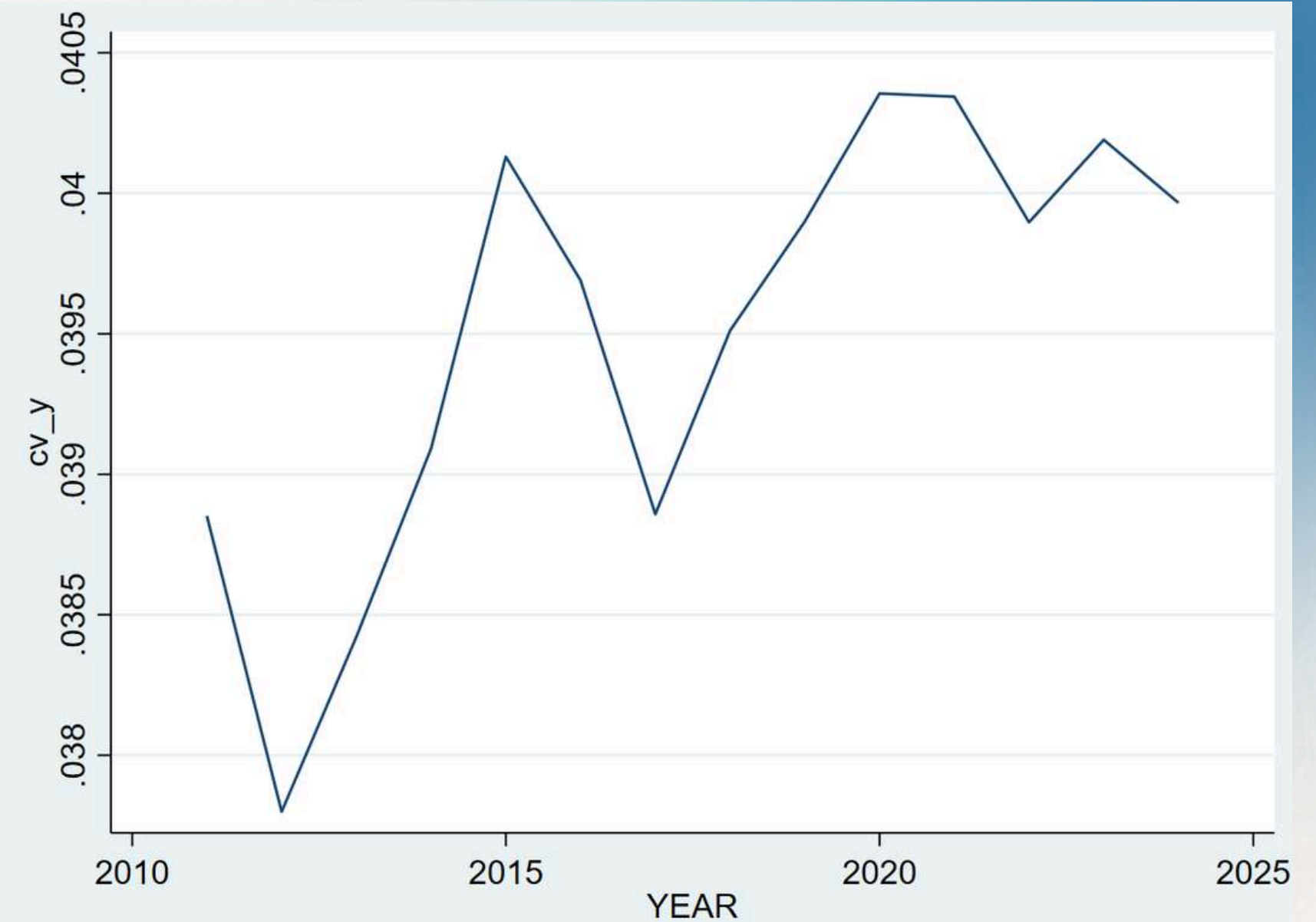
$$\sigma_t = \text{SD}(\ln y_{i,t})$$

$$\text{CV}_t = \text{SD}(y_t) / \text{mean}(y_t)$$



Sigma (σ) Convergence

- σ -divergence is occurring: interstate income inequality is widening, not narrowing
- This is evidence against σ -convergence among Indian states in this period



Robustness check

- Both graphs together strongly suggest no σ -convergence - richer states are pulling further ahead.

We define a gap variable $x(t)$ to track how far an economy is from its steady state.

$y(t)$: NSDP per capita

y^* : steady-state level of NSDP per capita

The speed at which the gap closes is proportional to the size of the gap

λ : speed of convergence

Negative sign : gap is shrinking

τ : length of time period where steady state doesn't change. Writing prior eqns at time $t-\tau$

- $x(t) = \ln y(t) - \ln y^*$

- **Dynamic equation**
 $dx(t)/dt = -\lambda x(t)$
 $x(t) = Ce^{-\lambda t}$ (after integration)

- $x(t-\tau) = \ln y_{i,t-\tau} - \ln y^*$

- $x(t) = x(t-\tau)e^{-\lambda\tau}$

$$\ln y_{i,t} = e^{-\lambda\tau} \ln y_{i,t-\tau} + (1 - e^{-\lambda\tau}) \ln y^*$$

$$\ln y_{i,t} = e^{-\lambda\tau} \ln y_{i,t-\tau} + (1 - e^{-\lambda\tau}) \ln y^* + \omega_{i,t}$$

In conditional convergence, steady state depends on structural variables. We have taken these:

- Human Development Index (HDI)
- Technology Index

$$\ln y_{i,t} = \alpha \ln y_{i,t-\tau} + \gamma_1 (\text{HDI}_{i,t}) + \gamma_2 (\text{Index}_{i,t}) + \omega_{i,t}$$

$$\tau = 1 \text{ Year}$$

LIMITATIONS OF USING OLS

01.

Dynamic Panel Bias

- OLS estimates become biased
- coefficients are inconsistent
- the bias is stronger when time periods are smaller

02.

Endogeneity Problem

This creates reverse causality, violating the OLS assumption:
 $\text{Cov}(X_{it}, u_{it}) = 0$

03.

Ignoring State-Specific Heterogeneity

States differ in different aspects.

OLS treats all states as identical and fails to control for these unobserved differences.

04.

Cross-Section Dependence

States may influence each other. OLS ignores such interactions, which may lead to misleading results.

MODEL USED: SYSTEM GMM

WHY SYSTEM GMM?

Handles Endogeneity

Uses internal instruments (lagged variables) to correct for reverse causality and omitted variable bias.

Suitable for Dynamic Panel Models

Correctly estimates models with lagged dependent variables without dynamic panel bias.

Controls for Unobserved Heterogeneity

Eliminates time-invariant state-specific effects through differencing.

Efficient for Small Time Periods and Many Cross-Sections

Works well for datasets with many states (N) and relatively few years (T).

Dynamic Panel Methodology

The growth model includes a lagged dependent variable:

$$y_{i,t} = \alpha y_{i,t-1} + \beta X_{i,t} + \mu_i + \varepsilon_{i,t}$$

where

$y_{i,t}$ = dependent variable
(income/output)

$X_{i,t}$ = explanatory variables

μ_i = unobserved state-specific effect

$\varepsilon_{i,t}$ = idiosyncratic error term

Removal of Fixed Effects

System GMM removes unobserved heterogeneity by first-differencing:

$$\Delta y_{i,t} = \alpha \Delta y_{i,t-1} + \beta \Delta X_{i,t} + \Delta \varepsilon_{i,t}$$

This eliminates the time-invariant state effect μ_i .

Instrumental Variables Approach

Lagged values of the variables are used as instruments to address endogeneity:

Instruments: $y_{i,t-2}$, $y_{i,t-3}$

These lags are assumed to be uncorrelated with the current error term.

System of Equations

System GMM estimates both the level equation and the differenced equation simultaneously, improving efficiency:

$$\begin{aligned} \Delta y_{i,t} &= \alpha \Delta y_{i,t-1} + \beta \Delta X_{i,t} + \Delta \varepsilon_{i,t} \\ y_{i,t} &= \alpha y_{i,t-1} + \beta X_{i,t} + \mu_i + \varepsilon_{i,t} \end{aligned}$$

PRE-ESTIMATION DIAGNOSTIC TESTS

- Panel Unit Root Test (Fisher-ADF Test) – to check stationarity of variables
- Multicollinearity Test (Variance Inflation Factor – VIF) – to detect correlation among regressors
- Heteroskedasticity Test (Breusch–Pagan Test) – to examine variance of error terms
- Serial Correlation Test (Wooldridge Test) – to detect autocorrelation in panel data

PANEL UNIT ROOT TEST (FISHER-ADF)

- Used to check stationarity of panel data variables
- Combines ADF tests across cross-sections

Augmented Dickey–Fuller (ADF) Test Equation

$$\Delta y_{i,t} = \alpha + \beta y_{i,t-1} + \sum \gamma_j \Delta y_{i,t-j} + \varepsilon_{i,t}$$

where

$\Delta y_{i,t}$ = first difference of the variable

$y_{i,t-1}$ = lagged level of the variable

γ_j = coefficients of lagged differences

$\varepsilon_{i,t}$ = error term

Hypothesis

$H_0: \beta = 0$ (Unit root present / non-stationary)

$H_1: \beta < 0$ (Series is stationary)

If the p-value < 0.05, we reject H_0

- Variables should be $I(0)$ or $I(1)$ before applying System GMM.

UNIT ROOT TEST OF $Y = \ln(\text{NSDP PERCAPITA})$

```
. xtunitroot fisher y, dfuller lags(1)
(831 missing values generated)
```

Fisher-type unit-root test for y
Based on augmented Dickey-Fuller tests

H0: All panels contain unit roots Number of panels = **12**
Ha: At least one panel is stationary Number of periods = **14**

AR parameter: **Panel-specific** Asymptotics: **T -> Infinity**
Panel means: **Included**
Time trend: **Not included**
Drift term: **Not included** ADF regressions: **1 lag**

		Statistic	p-value
Inverse chi-squared(24)	P	4.2304	1.0000
Inverse normal	Z	4.0204	1.0000
Inverse logit t(64)	L*	3.9238	0.9999
Modified inv. chi-squared	Pm	-2.8535	0.9978

P statistic requires number of panels to be finite.
Other statistics are suitable for finite or infinite number of panels.

```
. xtunitroot fisher D.y, dfuller lags(1)
(843 missing values generated)
```

Fisher-type unit-root test for $D.y$
Based on augmented Dickey-Fuller tests

H0: All panels contain unit roots Number of panels = **12**
Ha: At least one panel is stationary Number of periods = **13**

AR parameter: **Panel-specific** Asymptotics: **T -> Infinity**
Panel means: **Included**
Time trend: **Not included**
Drift term: **Not included** ADF regressions: **1 lag**

		Statistic	p-value
Inverse chi-squared(24)	P	44.4489	0.0068
Inverse normal	Z	-3.3058	0.0005
Inverse logit t(64)	L*	-3.1316	0.0013
Modified inv. chi-squared	Pm	2.9515	0.0016

P statistic requires number of panels to be finite.
Other statistics are suitable for finite or infinite number of panels.

Y is $I(1)$, suitable for System GMM.

UNIT ROOT TEST OF Y_{t-1}

```
. xtunitroot fisher L.y, dfuller lags(1)
(844 missing values generated)
```

Fisher-type unit-root test for $L.y$
Based on augmented Dickey-Fuller tests

H0: All panels contain unit roots	Number of panels =	12
Ha: At least one panel is stationary	Number of periods =	13

AR parameter:	Panel-specific	Asymptotics: T -> Infinity
Panel means:	Included	
Time trend:	Not included	
Drift term:	Not included	ADF regressions: 1 lag

		Statistic	p-value
Inverse chi-squared(24)	P	6.5675	0.9998
Inverse normal	Z	2.9669	0.9985
Inverse logit t(64)	L*	2.8046	0.9967
Modified inv. chi-squared	Pm	-2.5162	0.9941

P statistic requires number of panels to be finite.
Other statistics are suitable for finite or infinite number of panels.

```
. xtunitroot fisher D.L.y, dfuller lags(1)
(856 missing values generated)
```

Fisher-type unit-root test for **LD.y**
Based on augmented Dickey-Fuller tests

H0: All panels contain unit roots	Number of panels =	12
Ha: At least one panel is stationary	Number of periods =	12

AR parameter:	Panel-specific	Asymptotics: T -> Infinity
Panel means:	Included	
Time trend:	Not included	
Drift term:	Not included	ADF regressions: 1 lag

		Statistic	p-value
Inverse chi-squared(24)	P	43.2545	0.0093
Inverse normal	Z	-3.2380	0.0006
Inverse logit t(64)	L*	-3.0465	0.0017
Modified inv. chi-squared	Pm	2.7792	0.0027

P statistic requires number of panels to be finite.
Other statistics are suitable for finite or infinite number of panels.

Y_{t-1} is $I(1)$, suitable for System GMM.

UNIT ROOT TEST OF HDI

. xtunitroot fisher HDI, dfuller lags(1)			
(832 missing values generated)			
Fisher-type unit-root test for HDI			
Based on augmented Dickey-Fuller tests			
H0: All panels contain unit roots		Number of panels =	12
Ha: At least one panel is stationary		Number of periods =	14
AR parameter: Panel-specific		Asymptotics: T ->	Infinity
Panel means: Included			
Time trend: Not included			
Drift term: Not included		ADF regressions: 1	lag
		Statistic	p-value
Inverse chi-squared(24)	P	21.9447	0.5826
Inverse normal	Z	0.0612	0.5244
Inverse logit t(64)	L*	0.0302	0.5120
Modified inv. chi-squared	Pm	-0.2967	0.6166
P statistic requires number of panels to be finite.			
Other statistics are suitable for finite or infinite number of panels.			

. xtunitroot fisher dHDI, dfuller lags(1)			
(844 missing values generated)			
Fisher-type unit-root test for dHDI			
Based on augmented Dickey-Fuller tests			
H0: All panels contain unit roots		Number of panels =	12
Ha: At least one panel is stationary		Number of periods =	13
AR parameter: Panel-specific		Asymptotics: T -> Infinity	
Panel means: Included			
Time trend: Not included			
Drift term: Not included		ADF regressions: 1 lag	
		Statistic	p-value
Inverse chi-squared(24)	P	64.9498	0.0000
Inverse normal	Z	-5.0413	0.0000
Inverse logit t(64)	L*	-5.0221	0.0000
Modified inv. chi-squared	Pm	5.9106	0.0000
P statistic requires number of panels to be finite.			
Other statistics are suitable for finite or infinite number of panels.			

HDI is I(1), suitable for System GMM.

UNIT ROOT TEST OF TECHNOLOGY INDEX

```
. xtunitroot fisher index1, dfuller lags(1)
(832 missing values generated)
```

Fisher-type unit-root test for **index1**
Based on augmented Dickey-Fuller tests

H0: All panels contain unit roots Number of panels = **12**
Ha: At least one panel is stationary Number of periods = **14**

AR parameter: **Panel-specific** Asymptotics: **T -> Infinity**
Panel means: **Included**
Time trend: **Not included**
Drift term: **Not included** ADF regressions: **1 lag**

		Statistic	p-value
Inverse chi-squared(24)	P	17.3759	0.8322
Inverse normal	Z	1.4287	0.9234
Inverse logit t(64)	L*	1.6783	0.9509
Modified inv. chi-squared	Pm	-0.9561	0.8305

P statistic requires number of panels to be finite.
Other statistics are suitable for finite or infinite number of panels.

```
. xtunitroot fisher D.index1, dfuller lags(1)
(844 missing values generated)
```

Fisher-type unit-root test for **D.index1**
Based on augmented Dickey-Fuller tests

H0: All panels contain unit roots Number of panels = **12**
Ha: At least one panel is stationary Number of periods = **13**

AR parameter: **Panel-specific** Asymptotics: **T -> Infinity**
Panel means: **Included**
Time trend: **Not included**
Drift term: **Not included** ADF regressions: **1 lag**

		Statistic	p-value
Inverse chi-squared(24)	P	37.6843	0.0373
Inverse normal	Z	-1.7590	0.0393
Inverse logit t(64)	L*	-1.8668	0.0333
Modified inv. chi-squared	Pm	1.9752	0.0241

P statistic requires number of panels to be finite.
Other statistics are suitable for finite or infinite number of panels.

Technology Index is I(1), suitable for System GMM.

MULTICOLLINEARITY TEST (VIF)

- Check whether explanatory variables are highly correlated.
- High multicollinearity increases the variance of coefficient estimates and makes them unstable.

$$\text{VIF} = 1/(1 - R_i^2)$$

where

R_i^2 = Coefficient of determination from regressing X_i on other independent variables.

VIF < 5 → Acceptable

VIF ≥ 10 → Serious multicollinearity

MULTICOLLINEARITY TEST (VIF)

```
. vif
```

Variable	VIF	1/VIF
y		
L1.	3.69	0.270654
HDI	3.03	0.329517
index1	1.42	0.705375
Mean VIF	2.72	

VIF is 2.72, suitable for System GMM.

BREUSCH-PAGAN TEST (HETEROSKEDASTICITY)

- Tests whether error variance is constant.
- Correction not required beforehand → System GMM handles heteroskedasticity using robust standard errors.

Breush-Pagan Test Equation

$$u_i^2 = \alpha_0 + \alpha_1 X_{1i} + \alpha_2 X_{2i} + \dots + v_i$$

$$BP = nR^2$$

where:

n = number of observations

R^2 = coefficient of determination
from the auxiliary regression

$$BP \sim \chi^2(k)$$

k = number of independent variables

Hypothesis

H₀: Constant variance of errors
(Homoskedasticity)

H₁: Variance of errors depends on independent
variables (Heteroskedasticity present)

If the p-value < 0.05, we reject H₀

BP TEST

```
. estat hettest
```

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity

Assumption: Normal error terms

Variable: Fitted values of y

H0: Constant variance

chi2(1) = 2.30

Prob > chi2 = 0.1295

Since p-value > 0.05, we fail to reject H_0 .

No evidence of heteroskedasticity.

WOOLDRIDGE TEST (AUTOCORRELATION IN PANEL DATA)

- Tests for serial correlation in panel data errors.
- Pre-correction not required → System GMM handles autocorrelation internally.
- AR(1) and AR(2) tests are used to verify model validity.

Woolridge Test Equation

$$\Delta \varepsilon_{i,t} = \rho \varepsilon_{i,t-1} + u_{i,t}$$

where:

$\varepsilon_{i,t}$ = error term

ρ = autocorrelation coefficient

Hypothesis

H₀: No first-order autocorrelation ($\rho=0$)

H₁: First-order autocorrelation present ($\rho \neq 0$)

If p-value < 0.05, reject H₀

WOOLDRIDGE TEST

```
. xtserial y y1 HDI index1
```

Wooldridge test for autocorrelation in panel data

H0: no first-order autocorrelation

F(1, 11) = 9.454

Prob > F = 0.0106

Since p-value < 0.05, we reject H_0 .

First-order autocorrelation is present in the panel data.

STATA CODE FOR SYSTEM GMM ESTIMATION

```
xtabond2 y L.y HDI index1, gmm(L.y HDI index1, lag(2 3) collapse) twostep robust
```

- Lags (2-3) are used as instruments to address endogeneity and ensure valid, uncorrelated instruments in System GMM.
- Endogenous Variables: 1st lag of Y, HDI, Technology Index
- Uses two-step GMM estimator, which is more efficient.
- Provides heteroskedasticity-robust standard errors.

SYSTEM GMM RESULTS

```
. xtabond2 y L.y HDI index1 , gmm(L.y HDI index1 , lag(2 3) collapse) twostep robust
Favoring space over speed. To switch, type or click on mata: mata set matafavor speed, perm.
```

Dynamic panel-data estimation, two-step system GMM

Group variable: state_id	Number of obs	=	156
Time variable : YEAR	Number of groups	=	12
Number of instruments = 10	Obs per group: min	=	13
Wald chi2(3) = 628289.68	avg	=	13.00
Prob > chi2 = 0.000	max	=	13

y	Coefficient	Corrected std. err.	z	P> z	[95% conf. interval]	
y						
L1.	.9385457	.0374894	25.03	0.000	.8650678	1.012024
HDI	.7013428	.3407404	2.06	0.040	.0335039	1.369182
index1	.0361223	.0222861	1.62	0.105	-.0075577	.0798023
_cons	.2526959	.31989	0.79	0.430	-.3742769	.8796688

ANALYSIS OF ESTIMATION RESULTS

Wald $\chi^2(3) = 628289.68$

Prob > $\chi^2 = 0.000$

Interpretation: The model is jointly significant.

For β -convergence, the lagged y coefficient should be positive and $0 < \beta < 1$.

We obtained 0.93854.

This suggests conditional convergence across states.

A 1-unit increase in NSDP per Capita one period ago increases current NSDP Per Capita by 0.93 units on average.

POST-ESTIMATION TESTS IN SYSTEM GMM

- **Arellano–Bond Test** - Tests for autocorrelation in the differenced residuals.
- **Hansen / Sargan Test** - Tests the validity of instruments used in System GMM.

SERIAL CORRELATION TEST (ARELLANO-BOND)

- Checks for autocorrelation in the differenced residuals of the System GMM model.
- Tests Reported
 - AR(1) → First-order autocorrelation
 - AR(2) → Second-order autocorrelation
- Valid model: AR(1) significant, AR(2) insignificant.

Hypothesis

H₀: No serial correlation in residuals

H₁: Serial correlation present

If p-value < 0.05, reject H₀

Test Statistic

$$z = \hat{\rho} / SE(\hat{\rho})$$

where:

$\hat{\rho}$ = estimated autocorrelation coefficient of residuals

SE($\hat{\rho}$)= standard error of the estimated autocorrelation

$z \sim N(0,1)$

ARELLANO-BOND TEST

```
Arellano-Bond test for AR(1) in first differences: z = -2.63 Pr > z = 0.009  
Arellano-Bond test for AR(2) in first differences: z = 0.05 Pr > z = 0.959
```

AR(1) Test

Since $p\text{-value} < 0.05$, we reject H_0 .

This indicates first-order autocorrelation, which is expected in differenced residuals.

AR(2) Test

Since $p\text{-value} > 0.05$, we fail to reject H_0 .

This indicates no second-order autocorrelation.

AR(1) significant and AR(2) insignificant, confirming that the System GMM model is correctly specified and instruments are valid.

HANSEN / SARGAN TEST OF OVERIDENTIFYING RESTRICTIONS

- Tests the validity of instruments used in the System GMM model.
- Checks whether instruments are uncorrelated with the error term.

Test Statistic

$$\chi^2 = \hat{\mathbf{g}}' \mathbf{W}^{-1} \hat{\mathbf{g}}$$

Where:

$\hat{\mathbf{g}}$ = vector of sample moment conditions

$\mathbf{W}$ = weighting matrix

$$\chi^2 \sim \chi^2(\text{df})$$

Hypothesis

H₀: Instruments are valid (uncorrelated with error term)

H₁: Instruments are invalid

p-value > 0.05 → Fail to reject H₀

HANSEN / SARGAN TEST

Sargan test of overid. restrictions: $\chi^2(6) = 3.98$ Prob > $\chi^2 = 0.680$

(Not robust, but not weakened by many instruments.)

Hansen test of overid. restrictions: $\chi^2(6) = 7.26$ Prob > $\chi^2 = 0.297$

(Robust, but weakened by many instruments.)

Difference-in-Hansen tests of exogeneity of instrument subsets:

GMM instruments for levels

Hansen test excluding group: $\chi^2(3) = 2.79$ Prob > $\chi^2 = 0.425$

Difference (null H = exogenous): $\chi^2(3) = 4.47$ Prob > $\chi^2 = 0.215$

Sargan Test

Since $p\text{-value} < 0.05$, we reject H_0 .

This suggests possible instrument validity issues.

However, the Sargan test is not robust to heteroskedasticity, so it is less reliable in two-step GMM.

Hansen Test (Preferred Test)

Since $p\text{-value} > 0.05$, we fail to reject H_0 .

This indicates that the instruments used in the model are valid.

Difference-in-Hansen Test

- GMM Instruments for Levels
 $p\text{-value} > 0.05$

Instruments used in the levels equation are valid.

Speed of β -Convergence

The speed of convergence in β -convergence models is calculated as:

$$\lambda = -\ln(\alpha)/\tau$$

Where:

λ = speed of convergence

α = coefficient of the lagged dependent variable

τ = time interval

$$\lambda = 0.063$$

- The speed of convergence is 6.3% per year.
- This means that about 6.3% of the gap between current income and the steady-state level is closed each year.

Economic Implications of β -Convergence

- **Speed of Convergence**

The estimated $\alpha = 0.93$ implies a convergence speed (λ) of about 6% per 1-year period.
However, convergence is not immediate due to economic inertia.

- **Slow Speed of Convergence ($\approx 6.3\%$)**

The estimated convergence speed suggests that only about 6.3% of the income gap closes each year
Indicates that convergence is slow and long-term in nature
Regional inequality may persist for a long period despite convergence

- **Role of HDI**

HDI is statistically significant, confirming it has a real impact on income.
It shows that higher human development significantly increases economic growth and supports conditional convergence across states.

- **Policy Implications**

Convergence is not automatic, requires policy support

Key policy focus:

- Investment in education and healthcare
- Development of infrastructure and technology
- Targeted support for backward states

Promotes balanced and inclusive growth across regions

PHILLIPS & SUL (2007) CLUB CONVERGENCE METHOD

- Tests whether economies converge as a whole or form sub-groups (clubs)
- Allows for heterogeneity across states
- Express each state relative to cross-sectional average
- Tracks how each state evolves over time compared to others

Core Test: Log-t Regression

Uses cross-sectional variance of relative transitions

Regression coefficient determines convergence

Decision Rule:

Coefficient $\geq -2 \rightarrow$ Convergence

Coefficient $< -2 \rightarrow$ No convergence

If No Overall Convergence:

Apply clustering algorithm

Identify convergence clubs

Results of Club Convergence (Log-t Test)

```
. bysort YEAR: egen mean_y = mean(y)
(832 missing values generated)

. gen h_it = y / mean_y
(832 missing values generated)

. bysort YEAR: egen sd_h = sd(h_it)
(832 missing values generated)

. gen H_t2 = sd_h^2
(832 missing values generated)

. gen log_H = ln(H_t2)
(832 missing values generated)

. gen log_t = ln(YEAR)
(832 missing values generated)
```

```
. gen log_t = ln(YEAR)
(832 missing values generated)
```

```
. reg log_H log_t
```

Source	SS	df	MS	Number of obs	=	168
Model	.145839559	1	.145839559	F(1, 166)	=	236.58
Residual	.102331772	166	.000616456	Prob > F	=	0.0000
				R-squared	=	0.5877
				Adj R-squared	=	0.5852
Total	.248171331	167	.001486056	Root MSE	=	.02483

log_H	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
log_t	14.74593	.9587065	15.38	0.000	12.8531	16.63876
_cons	-118.6741	7.295385	-16.27	0.000	-133.0778	-104.2704

- The log-t test coefficient (14.75) is greater than -2, indicating strong convergence among states.
- Indicates no need to form separate convergence clubs.

DO POORER INDIAN STATES GROW FASTER?

- **Yes, evidence of conditional β -convergence.**

Poorer states grow faster after controlling for factors like HDI.

- **However, σ -divergence is observed.**

Income inequality across states is still increasing.

- **Convergence is slow (~6.3% per year)**

Catch-up exists but is too gradual to reduce inequality gap.

- **No club convergence found**

No club convergence is detected, though states may differ in their long-run steady state.

- **Human development (HDI) is crucial**

States with better HDI grow faster, shaping convergence outcomes.

LIMITATIONS OF THE ANALYSIS

Limited Variable Selection

Key growth determinants like institutions, policy quality, inequality, and governance are not included.
May lead to omitted variable bias.

Proxy-Based Measurement

Technology Index constructed using limited proxies (telephone, bank ratio, power).
May not fully capture true technological progress.

Cross-Sectional Dependence Not Tested

States may influence each other (spillover effects).
Ignoring this can lead to biased and inconsistent estimates.

Ignoring Structural Breaks

Events like policy changes, COVID-19, economic shocks not explicitly modeled.
Can affect growth dynamics.

Ignoring Informal Economy

Large informal sector in India not captured in NSDP.
May underestimate true economic activity.

**THANK YOU
VERY MUCH**

**KHWAHISH
AVNI
PALLVI
YOGYANSH**